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Pixelation-Free, Monolithic Iontronic Pressure Sensors Enabling Large-Area Simultaneous Pressure and Position Recognition via Machine Learning

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Abstract

Despite significant advances, flexible pressure sensors still face critical limitations, including pixelated architecture and extensive wiring, which hinder their scalability for large-area applications. Here, we present a pixelation-free, monolithic iontronic pressure sensor that simultaneously detects pressure and position over a large area with high sensitivity (4.16 kPa^{-1}), a wide detection range (<455 kPa), and excellent durability. The device, consisting of a single ionogel film bridging multiple peripheral electrodes, eliminates the structural complexity of conventional designs. Under AC bias, mechanical pressure releases ions from polymer chains inside the ionogel, increasing ionic conductivity and establishing spatially varying impedance pathways to each electrode. This enables the sensor to capture both spatial and pressure information in the output current. Continuous ion oscillation under AC bias facilitates sustained signal generation under static pressure (>3300 s) without significant degradation, thereby overcoming key limitations of conventional piezoionic pressure sensors. Machine learning algorithms effectively decouple overlapping pressure–position signals from the multichannel outputs, achieving high accuracy in pressure prediction across different spatial positions. Real-time demonstrations, including handwriting recognition with pressure-dependent color variation and simultaneous multi-point detection producing piano sounds of varying tones, highlight the versatility of the device for next-generation large-area and flexible human–machine interfaces.

Juhui Kim, Junseo Lee, and Kwang-Hun Choi contributed equally to this work.

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1 | Introduction

Flexible pressure sensors capable of detecting subtle pressure variations over large, deformable surfaces have emerged as a cornerstone technology, accelerating recent advances in human–machine interfacing (HMI), including wearable electronics, soft robotics, and electronic skin (e-skin) [1, 2]. This progress has been driven by extensive efforts to develop soft and highly mechanosensitive electronic materials. Representative strategies involve elastomer composites incorporating nanofillers such as carbon nanotubes, silver nanowires, and piezoelectric ceramics like PZT or $BaTiO_3$, which enable flexible sensors with high sensitivity comparable to human skin [3–6]. Typically, conventional flexible pressure sensors have been realized through three signal transduction mechanisms: piezoresistive, which detects applied force via changes in resistance; piezoelectric, which generates transient charges in response to deformation; and piezocapacitive, which monitors variations in capacitance under load [7, 8]. From a perspective of device integration, these conventional devices commonly adopt a vertically stacked sandwich architecture, in which a flexible piezoresponsive material is positioned between two electrodes to form effective electric fields and conduction paths (Figure 1a). In particular, to obtain spatial information of pressing contact, these devices require pixelated arrays, which further increase fabrication challenges. This design also significantly increases the number of interconnects (i.e., number of pixels x 2) and wiring complexity, thereby limiting scalability to large areas. Furthermore, the repeated pressure stimuli during real-world operation can also degrade electrode–active layer interfaces, compromising long-term durability. Therefore, despite remarkable progress in mechanosensitive materials and device sensing performance, achieving high-resolution, robust, and scalable large-area pressure sensing remains a formidable challenge, highlighting the critical need for innovative design strategies beyond conventional architectures.

Recently, mechanosensitive piezoionic gels such as ionic hydrogels and ionogels have attracted considerable attention as promising candidates for flexible pressure sensors [9, 10]. Their mechanosensitive response arises from piezoionic dynamics, in which mechanical deformation redistributes mobile ions and generates electrical signals without the need for an external bias [11]. This self-powered mechanism is conceptually analogous to piezoelectric sensing, except that ions—rather than electrons or holes—serve as the primary charge carriers [12]. Owing to their superior viscoelasticity, biocompatibility, and self-healing capabilities, the piezoionic gels have also been successfully applied to diverse sensing applications, including self-powered strain sensors [13] and bionic mechanoreceptors [9], stretchable multimodal [14] or self-healable [15] tactile sensors. However, piezoionic signals are inherently transient because they appear only at the moment of pressure application or release; once the ions reach a new equilibrium, the output quickly vanishes, making static-pressure sensing difficult.

To overcome the transient nature of self-powered sensing, several studies applied external bias to operate the piezoionc gels in an iontronic mode, where pressure-induced changes in ionic impedance provide a continuous and tunable electrical readout. This approach improves signal stability and enables more quantitative pressure sensing compared with purely self-powered operation. Nevertheless, most piezoionic gel-based iontronic pressure sensors still rely on DC bias operation, where the pressure-induced current or voltage spike again decays toward equilibrium, limiting long-duration static-pressure detection. For instance, the maximum detectable static pressure duration has been limited to 100 s with 30% signal degradation [14]. Moreover, obtaining spatial information still requires arrays of discrete sensing units, thereby retaining the fundamental challenges of wiring complexity and circuit integration for large-area scalability. Additionally, the development of piezoionic-gel based iontronic pressure sensor systems capable of operating durably and with environmental stability over large pressure ranges remains a critical requirement for practical implementation.

Herein, we suggest a pixelation-free monolithic iontronic pressure sensor (PF-MPS), operated under an AC bias, in which a single ionogel layer bridges multiple peripheral electrodes, as illustrated in Figure 1b. The ionogel layer serves as a scalable, mechanically flexible, and conformable sensing medium, eliminating the discontinuities inherent in pixelated designs (Figure 1c). This monolithic architecture dramatically reduces interconnect complexity—requiring only four electrodes regardless of sensing area size—while also enhancing robustness by removing vulnerable electrode interfaces. Pressure applied at any location on the ionogel induces a local redistribution of ions, which is simultaneously recorded from different channels as current signals, enabling both pressure magnitude and position information without the need for discrete pixels. To further separate the overlapping contributions of pressure and position, the outputs from the four electrodes are processed through a machine-learning framework (Figure 1d), which decouples and reconstructs both features with high accuracy. This strategy achieves high spatial resolution without relying on dense pixelation. Moreover, the machine-learning algorithms can effectively handle the intertwined pressure–position signals that would be difficult to deconvolute using conventional approaches.

2 | Results and Discussion

As a pressure-sensitive, piezoionic material, we selected a poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP)–based ionogel incorporating 1-ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide ([EMIM][TFSI]) ionic liquid for its high ionic conductivity, exceptional mechanical robustness, flexibility, and electrochemical stability [16–18]. The PVDF-HFP ionogel films were fabricated via a simple solution casting process, in which a homogeneous blend of PVDF-HFP and [EMIM][TFSI] dissolved in acetone was cast into a mold and the solvent evaporated under ambient conditions. This facile and highly scalable method enables the production of large-area ionogel films without specialized equipment or controlled atmospheres. As shown in Figure 1c, ionogel films with diameters up to 11 cm were readily fabricated, exhibiting uniform planarity and high flexibility.

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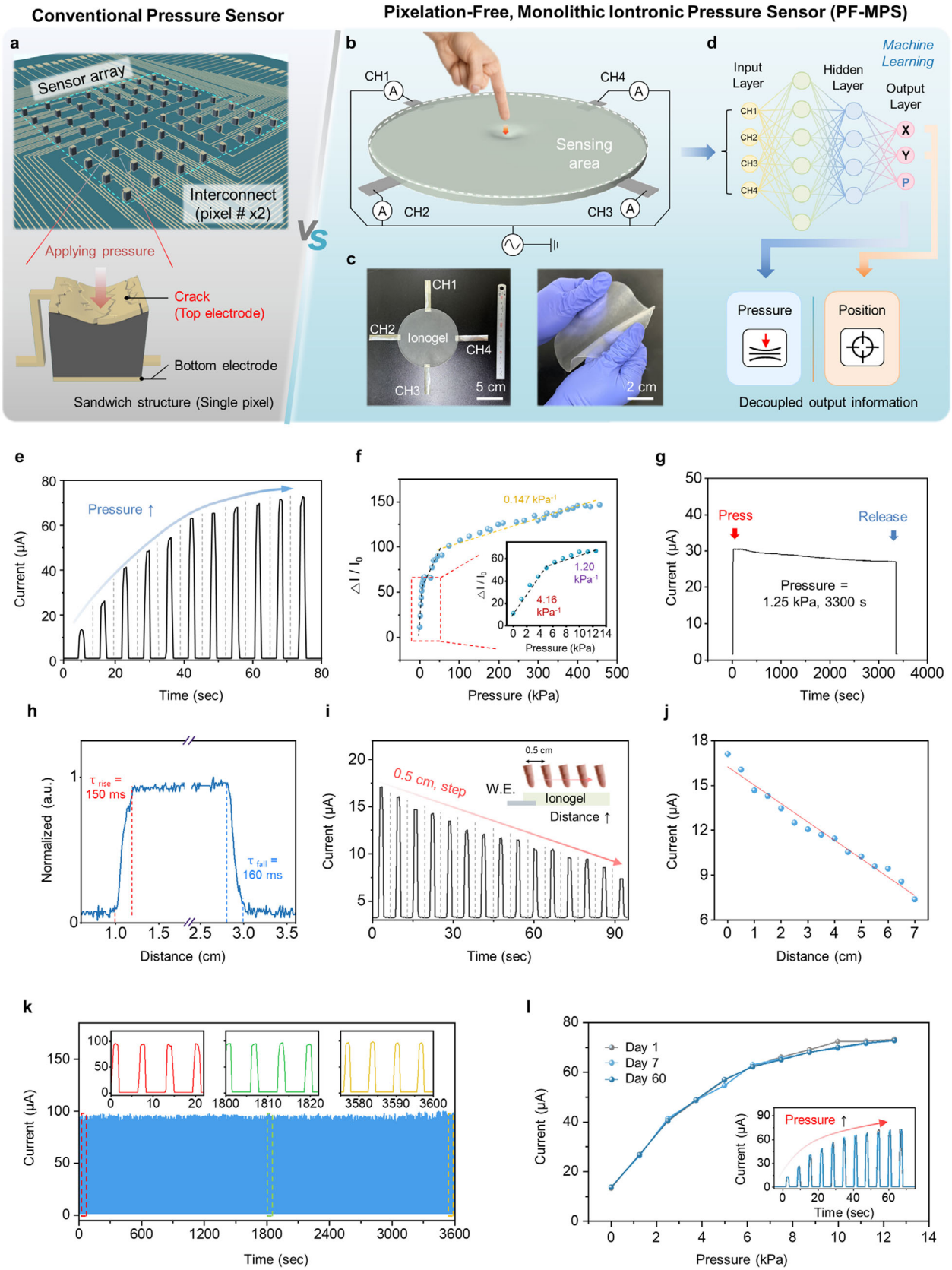

FIGURE 1 | Design and sensing performance of pixelation-free, monolithic iontronic pressure sensor (PF-MPS). (a) Schematic illustration of the limitations of conventional pressure sensors based on sensor arrays with sandwich electrode structures. (b) Schematic illustration of the PF-MPS with the corresponding circuit diagram for the sensing system. (c) Photographs of the fabricated ionogel-based PF-MPS device, showing its dimensions and mechanical flexibility. (d) Diagram of the decoupling strategy for pressure and position sensing using machine learning. (e) Current response of the PF-MPS under sequential pressure loading (from 0 to 12.5 kPa, in 1.25 kPa increments) under an applied AC bias of 2 Vpp at 5 kHz. (f) Calculated pressure-dependent current sensitivity in the range of 0 to 455 kPa. The inset shows a magnified view of the low-pressure region (0–12.5 kPa).

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Figure 1e exhibits the pressure-dependent current responses of the PF-MPS based on a PVDF-HFP ionogel film containing 20 wt.% [EMIM][TFSI], measured under an AC bias of ± 1 V (peak-to-peak, Vpp) at 5 kHz. As the applied pressure was gradually increased from 0 to 12.5 kPa in increments of approximately 1.25 kPa, the measured current correspondingly increased from 13.8 to 72.8 μA, indicating pressure-sensing capability. To further assess this response quantitatively, the pressure sensitivity (S) was evaluated using the following equation:

$$S = \frac{\Delta I / I_0}{\Delta P}$$

where I_0 is the initial current in the unloaded state, ΔI is the current difference between the unloaded state and a given applied pressure, and ΔP is the corresponding pressure difference [19]. This formulation provides a normalized assessment of sensitivity across different loading conditions. The PF-MPS exhibited sensing ability up to ~455 kPa (Figure 1f), confirming an exceptionally wide pressure range. Furthermore, the PF-MPS exhibits a high sensitivity of 4.16 kPa^{-1} in the low-pressure regime (0–6.24 kPa) and 1.76 kPa^{-1} in the higher range (6.24–12.5 kPa). These results confirm that the PF-MPS exhibits sensitivity and pressure range comparable to various simultaneous pressure/position sensing systems, while maintaining robust performance across both low and high pressure regimes (Table S1 and Figure S1).

Unlike conventional DC-driven iontronic sensors exhibiting transient current spikes upon pressure loading, the PF-MPS operating under an AC bias can reliably detect static pressure. Figure 1g shows that only ~10% degradation in the output current was observed even under prolonged loading for over 3300 s. The output current was maintained without noticeable decrease when a constant 1.25 kPa load was applied for over 100 s (Figure S2). This sustained signal might be attributed to ion oscillation within the ionogel under AC bias, where pressure-induced impedance changes are continuously reflected in the output current. A detailed discussion of the device's operational mechanism is provided in the following section. In addition, the PF-MPS exhibited fast response behavior, with rise and fall times of 160 and 150 ms, respectively (Figure 1h). The measured response time reflects the intrinsic ionic response of the ionogel-based PF-MPS device, as it was obtained prior to any circuit-level processing or machine-learning inference. These response times are well within the time scale of human touch detection, highlighting the potential of the PF-MPS for real-time interactive HMI applications.

The current response of the AC-driven PF-MPS is governed by the distance-dependent ionic impedance of the pathway formed between the contact point and the electrodes. As shown in Figure 1i, a constant pressure of 2.5 kPa was applied to an 11 cm diameter PF-MPS, and the contact point was sequentially shifted away from a readout electrode in 0.5 cm increments. As the distance from the electrode increased, the measured current linearly decreased (Figure 1j), directly reflecting the increased ionic impedance along the ionogel conduction path. These results demonstrate that both the pressure and the spatial location of the contact point can be simultaneously extracted from the current response of the PF-MPS. The methodology for decoupling this inherently coupled information through machine-learning analysis will be discussed in a subsequent section.

Furthermore, the long-term reliability of the PF-MPS was evaluated by examining its durability and operational stability under repetitive loading and extended storage. As shown in Figure 1k, the PF-MPS maintained consistent current responses during 560 pressure cycles (10 kPa, 1 s per cycle) spanning a total duration of 3600 s, confirming its mechanical robustness and signal reproducibility. In addition, we evaluated its stability under extended storage conditions, in which the samples were kept in covered Petri dishes under ambient atmosphere at 23–26°C and approximately 60% relative humidity, without intentional light exposure. As evidenced by nearly identical current-pressure profiles before and after storage, no significant degradation in sensing performance was observed even after 60 days of ambient storage (Figure 1l). These results highlight the stability of the ionogel-based PF-MPS, successfully addressing the inherent limitations of hydrogel-based sensors, which are prone to dehydration and consequent performance degradation over time. Previously, the AC-driven, large-area ionic hydrogel touch panel that can detect spatial information without the need for pixelation and static touch time by using changes in capacitance has been demonstrated; however, this device did not respond to the pressure changes [20]. To the best of our knowledge, the PF-MPS is the first demonstration of a pixelation-free monolithic iontronic flexible pressure sensor that simultaneously detects pressure and position over a large area. The device maintains high sensitivity, long static-pressure sensing capability, and excellent durability, enabled by a minimal four-electrode architecture that eliminates the wiring complexity of pixelated sensor arrays (Table S1 and Figure S1).

To gain mechanistic insight into the sensing characteristics of the PF-MPS, we investigated the ionic displacement behavior within the ionogel under mechanical pressing. We first examined the pressure-induced microstructural changes in the ionogel using 2D X-ray diffraction (2D-XRD). As shown in Figure 2a, in the unpressed state, two distinct diffraction peaks were observed at q = 1.3 and 1.4 $Å^{-1}$, corresponding to the (100) and (020) reflections of the PVDF-HFP matrix [21]. However, when the ionogel film was vertically pressed, as shown in Figure S3, a new diffraction peak emerged near q = 0.41 $Å^{-1}$, which was absent in the unpressed condition. The original diffraction peaks of the PVDF-HFP matrix remained unchanged after pressing, indicating that the polymer crystalline framework was preserved. These pressure-induced variations in diffraction features are more clearly revealed in the 1D line-cut profiles in Figure 2b. Consistently, the same trend appears across all IL contents

(g) Long-term DC current response of the PF-MPS under a static pressure of 1.25 kPa maintained for over 3000 s. (h) Response time of the PF-MPS, showing rise and fall times. (i) Current response as a function of the distance between the electrode and pressing position. (j) Linear decrease of the current response with increasing electrode-to-pressing position distance. (k) Cyclic stability of the PF-MPS under repeated loading at 10 kPa for 3600 s. (l) Long-term durability and reproducibility of the PF-MPS performance over 60 days.

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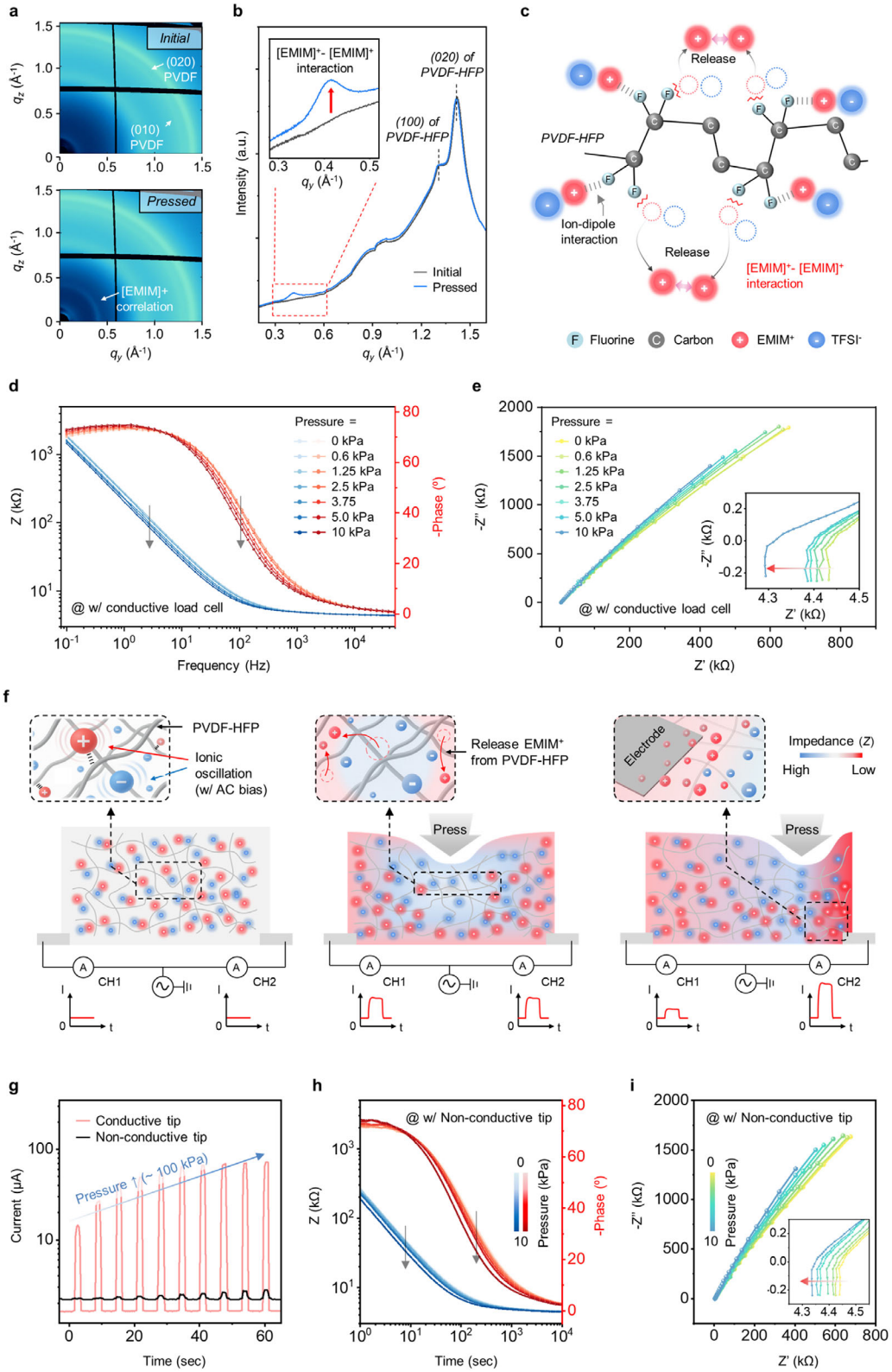

FIGURE 2 | Working principle and operation mechanism of ionogel and based PF-MPS under mechanical stimulation. (a) 2D synchrotron XRD patterns of the ionogel film in the initial and pressed states. (b) Corresponding 1D line-cut profiles of XRD patterns, highlighting the revealed $[EMIM]^+$–$[EMIM]^+$ correlations peaks at the pressed state. (c) Schematic illustration of the ion displacement behavior (releasing of EMIM+ cations) in ionogel under an applied pressure. (d) Bode plots and (e) Nyquist plots from electrochemical impedance spectroscopy (EIS) measured under different applied pressures. (f) Schematic representation of the sensing mechanism in PF-MPS, where AC bias drives ion oscillation and mechanical pressure releases cations from the polymer matrix, leading to impedance changes. (g) Comparison the current response under gradually increasing pressure applied with a plastic (non-conductive, black) and metal (conductive, light red) tips (ranging from 10 to 100 kPa (10 kPa increments). (h) Bode plot and (i) Nyquist plot obtained from EIS measurements with a plastic (non-conductive) loading tip by varying the pressure from 0 to 10 kPa.

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(10–40 wt.%), as shown in Figure S4. The newly appeared peak is assigned to ion–ion correlations between $[EMIM]^+$ cations that become mobile upon release from the PVDF-HFP matrix [22, 23].

The illustration in Figure 2c schematically depicts the ionic displacement behavior within the ionogel under applied pressure. In the unpressed state, $[EMIM]^+$ cations are predominantly coordinated to the fluorine end groups of the PVDF-HFP polymer chains through ion–dipole interactions [22]. This coordination suppresses the $[EMIM]^+$–$[EMIM]^+$ correlations typically observed in the neat ionic liquid [EMIM][TFSI], leading to the absence of the corresponding diffraction peak in the ionogel [22, 24]. However, upon mechanical deformation, a distinct diffraction peak corresponding to $[EMIM]^+$–$[EMIM]^+$ interactions energies, indicating that the applied pressure disrupts the ion–dipole interactions between $[EMIM]^+$ cations and the PVDF-HFP matrix. Consequently, mobile ions are released within the polymer matrix and can interact with one another [25, 26, 27]. This ion release mechanism is further facilitated by the inherent microstructural characteristics of the PVDF-HFP ionogel. As revealed by scanning electron microscopy (SEM) imaging (Figure S5), the ionogel exhibits a porous polymer network microstructure, which may contribute to localized stress concentration in the contact area, thereby enhancing the detachment of $[EMIM]^+$ cations from the PVDF-HFP matrix.

To understand how these liberated cations affect PF-MPS performance, we performed electrochemical impedance spectroscopy (EIS) under AC bias to analyze pressure-dependent behavior. As shown in the Bode plots in Figure 2d, the impedance (Z) gradually decreased, with a reduction of approximately 27.4% as the applied pressure on the ionogel film increased. In addition, the phase angle shifted from higher values (capacitive) toward lower values approaching 0° across the frequency range of 5.31 Hz to 5.89 kHz, indicating a transition toward resistive dominance in the ionogel. According to the complex impedance equation,

$$Z = R + \frac{1}{j\omega c}$$

where, R represents the resistive (real) component arising from bulk ionic conduction, and C denotes the capacitive (imaginary) component associated with ion–dipole interactions. Therefore, the observed phase shift indicates a reduced capacitive contribution and an increased relative contribution from resistive ion transport [28, 29]. More specifically, the electrical response of the sensor transitions from a charge-storage-dominated regime to one characterized by more immediate current conduction, resulting in a reduced phase lag between voltage and current. These findings indicate that the number of liberated $[EMIM]^+$ cations in the ionogel increases with applied pressure, leading to a reduction in bulk impedance.

These findings are further supported by the Nyquist plots shown in Figure 2e. The absence of a distinguishable semicircle across the high- to mid-frequency range indicates that the ionogel exhibits predominantly resistive behavior and that interfacial charge storage at the electrodes is not a major contributor to the overall impedance. As the applied pressure increased, the real-axis intercept (Z′) decreased. According to the following equation, this indicates a reduction in bulk resistance (R_b) and, consequently, an increase in ionic conductivity (σ). According to the relationship:

$$\sigma = \frac{l}{R_b A} = \sigma\ \mu_i q_i\ c_i = \frac{q_i^2 c_i D_i}{kT}$$

where σ is ionic conductivity, l is the ion transport distance, A is the electrode area, μ_i is the ion mobility, q_i is the ionic charge, c_i is the mobile ion concentration, D_i is the diffusion coefficient, k is the Boltzmann constant, and T is the absolute temperature [30, 31]. The reduction in R_b with increasing pressure confirms that ionic conductivity is enhanced due to a greater population of mobile ions released from ion–dipole interactions with the PVDF-HFP matrix [31, 32]. These results demonstrate that signal generation in the PF-MPS originates from pressure-induced ion release and redistribution of $[EMIM]^+$, which reduces bulk resistance and suppresses capacitive effects, thereby facilitating more efficient ion transport under the applied AC bias. To further verify the reliability of this mechanism, we conducted repeated loading–unloading EIS measurements under high pressure (354 kPa) (Figure S6). The device consistently showed reduced impedance and bulk resistance at high pressure and fully recovered to its initial state at 0 kPa, with this reversible behavior maintained over five cycles.

The working principle of the PF-MPS is schematically illustrated in Figure 2f. Under AC bias, ions within the ionogel oscillate periodically in response to the alternating electric field, producing a baseline current. When mechanical pressure is applied, ions previously bound to the polymer chains through ion–dipole interactions are released, thereby increasing the population of mobile ions. This release enhances the overall ionic conductivity and leads to a greater current response. In addition, pressing the ionogel at a specific location creates a localized potential difference between the pressing point and each electrode, establishing spatially varying impedance pathways across the gel. Electrodes situated closer to the pressing point encounter lower impedance due to shorter conduction paths and higher local ion density, thus delivering stronger current signals. This mechanism enables simultaneous detection of both pressure magnitude and spatial location using only four lateral electrodes. This AC-driven iontronic response is fundamentally different from that observed under conventional DC-bias operation. As further confirmed by the DC control experiment in Figure S7, a constant bias drives ions unidirectionally and rapidly dissipates the pressure-induced ionic gradient, producing only transient current spikes rather than sustained signals.

Considering the underlying operating mechanism of the device, signal generation under AC bias can be influenced by the electrical (e.g., capacitive) coupling characteristics of the contacting material. To investigate this effect, we examined the device response with different contact materials. Indeed, contact with conductive materials yielded significantly higher current responses, which can be attributed to enhanced capacitive coupling between the conductive contact and the ionogel surface under AC bias [33, 34]. However, more importantly, the PF-MPS maintained a distinct pressure-sensitive response even when contacted with non-conductive materials (e.g., a plastic loading tip), as shown in Figure 2g. More specifically, pressure-dependent EIS measurements using a plastic loading tip (Figure 2h,i)

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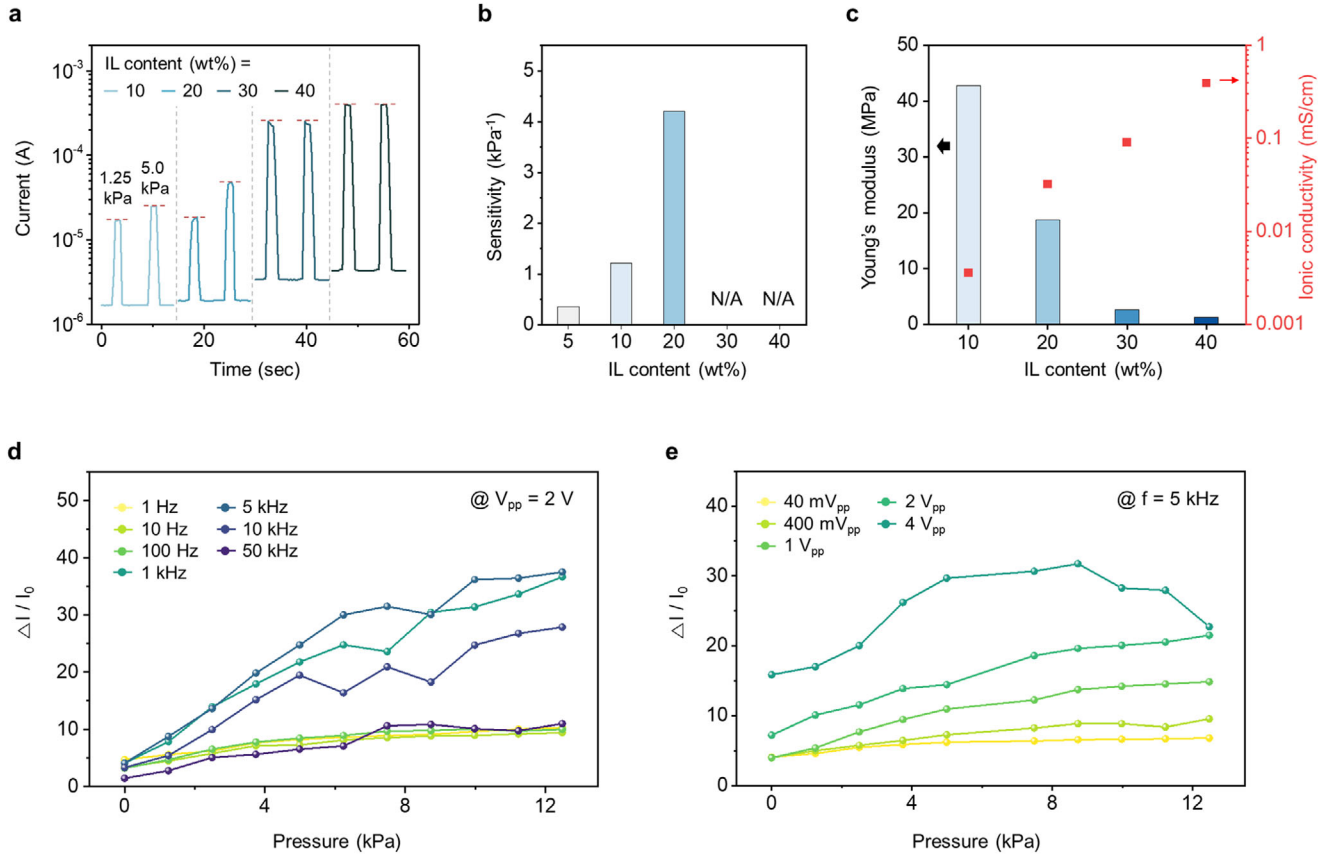

FIGURE 3 | Influence of ionic liquid (IL) content in ionogel and electrical input conditions on PF-MPS performance. (a) Current responses of the PF-MPS at 1.25 and 5.0 kPa with varying IL contents (10–40 wt.%). b) Calculated sensitivity and (c) Young's modulus and ionic conductivity, as a function of IL content in ionogel. (d) Applied bias frequency-dependent sensitivity of PF-MPS, applied bias amplitude was fixed as 2 Vpp across a wide frequency range (1 Hz to 50 kHz). (e) Applied bias amplitude-dependent sensitivity of PF-MPS measured at 5 kHz under varying AC bias amplitudes (40 mVpp–4 Vpp).

exhibited changes in impedance and phase angles similar to those obtained with the stainless steel (SUS) loading tip shown in Figure 2d,e. Additionally, comparative tests revealed that the device exhibited distinct pressure-responsive current changes regardless of whether it was placed on a conductive substrate (copper plate) or an insulating substrate (glass slide) (Figure S8). These results conclusively support that the pressure-induced current in the PF-MPS is governed primarily by piezoionic effects within the ionogel film rather than by the external capacitive coupling effect.

Since the ionic conductivity and impedance of the ionogel film in PF-MPS are strongly influenced by ionic liquid (IL) content, we investigated the pressure sensing behavior of the device using the ionogels with varying IL concentrations (5–40 wt.%). As shown in Figure 3a, ionogels with 10 and 20 wt.% IL exhibited a pronounced increase in current when the applied pressure increased from 1.25 to 5 kPa. In contrast, the 30 and 40 wt.% IL ionogels showed no response to pressure changes, although higher baseline current and contact-induced responses were observed due to enhanced ionic conductivity and capacitive coupling, respectively. This behavior is clearly reflected in the sensitivity analysis, as shown in Figure 3b. The pressure sensitivity of the PF-MPS improved with increasing ionogel IL content up to 20 wt.%, beyond which the pressure-dependent modulation diminished. This optimal ionogel composition is closely related to the competing effects of mechanical modulus and ionic conductivity changes with varying IL contents. As shown in Figure 3c, Young's modulus of the ionogels decreased markedly from 42.85to 1.25 kPa, whereas the ionic conductivity continuously increased with increasing IL content from 10 to 40 wt.%. The optimal Young's modulus and ionic conductivity obtained from the 20 wt% IL ionogel were approximately 18.8 MPa and 0.09 mS/cm, respectively, which represent a higher modulus and ion conductivity compared to those of previously reported piezoionic pressure sensors with ionogels [13, 35]. At low IL contents, the films remain relatively stiff, limiting strain-induced ion release, whereas at high IL contents, they become overly soft and already contain abundant mobile ions, such that the additional current change from strain-induced ion release becomes negligible. The cyclic stress–strain curves indicate that the 20 wt.% IL ionogel maintains elastic recovery up to ~20% strain, exhibiting reversible deformation without significant permanent set (Figure S9b). Therefore, the ionogel containing 20 wt.% IL provides an optimal balance, offering sufficient softness for effective ion release while retaining mechanical integrity, thereby enabling the highest pressure-sensitivity in PF-MPS.

Furthermore, to identify optimal electrical operating parameters, we systematically varied the frequency and amplitude of the

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applied AC voltage while quantitatively evaluating the pressure-sensing performance. Figure 3d presents the calculated sensitivity of the device at different frequencies under a fixed amplitude of 2 Vpp. As the frequency increased from 1 Hz to 5 kHz, both the baseline current and the pressure-induced current change progressively increased, resulting in a corresponding rise in sensitivity (Figure S10). This trend is attributed to the frequency-dependent decrease in system impedance, which facilitates enhanced ionic oscillation and displacement. The maximum sensitivity was observed at 5 kHz, indicating the most efficient ion response and pressure detection performance. However, beyond this point, the response became unstable, and the sensitivity decreased, likely due to the limited ability of ions to follow rapid voltage oscillations. In addition, the applied voltage amplitude also affected device performance (Figure 3e). At 2 Vpp, the PF-MPS maintained a stable, high sensitivity with linearly increasing current responses over the applied pressure range, whereas higher amplitudes (e.g., 4 Vpp) induced excessive ion oscillation that diminished the relative contribution of pressure-driven ion redistribution, ultimately reducing signal fidelity. Based on these results, we identified 5 kHz and 2 Vpp as the optimal AC driving conditions for maximizing piezoionic sensing in PF-MPS, providing a robust strategy for high-performance and reliable pressure detection.

The fundamental challenge in position and pressure sensing lies in extracting both spatial and force information from limited sensor outputs while maintaining system simplicity and computational efficiency. Figure 4a illustrates the four-channel sensing system configuration, including PF-MPS, and an interface board for current-to-voltage conversion and signal processing. When an AC signal is applied to the four electrodes, mechanical stimulus to different regions varies impedances between the stimulated location and each electrode, resulting in distinct current distributions across the four channels. These currents are converted to voltage signals through 100Ω shunt resistors, subsequently amplified using INA185 operational amplifiers, and processed through AD8436 analog root mean square (RMS)-to-DC converters before digital acquisition by an Arduino Mega microcontroller. The implementation of analog RMS-to-DC conversion provides significant advantages over direct AC sampling approaches, as it reduces the required MCU sampling speed and eliminates the computational burden of real-time RMS calculations, thereby enabling more efficient signal processing. Representative measurement is shown in Figure 4b, demonstrating the spatial discrimination capability of the system. Position (3,2) in a 3 × 3 grid was subjected to mechanical stress with varying pressures from 0 kPa to 12.5 kPa in eleven discrete pressure levels. The voltage response patterns clearly reflect the geometric relationship between the stimulated location and electrode positions: CH2 (green) exhibits the highest voltage amplitude due to its proximity to position (3, 2), while CH1 (blue) and CH3 (red) show nearly identical responses corresponding to their equidistant placement from the stimulated location. CH4 (purple) registers the lowest voltage, consistent with its maximum distance from the stimulation point. The distance-dependent voltage distribution forms the foundation for position recognition. More comprehensively, the voltage intensity mapping across all nine grid regions is presented in Figure 4c, revealing the systematic relationship between electrode proximity and signal strength. For each electrode, the voltage amplitude decreases proportionally with distance from the stimulated location, creating unique four-channel signatures for each spatial position. These spatial sensitivity patterns remained consistent even under mechanical deformation, as validated by reliable operation on a cylindrical substrate (Figure S11).

To quantitatively assess the spatial resolution of the PF-MPS, we conducted a systematic evaluation along a 60 mm linear trajectory at an applied pressure of 406 kPa. A load cell pressed the sensor at 601 discrete positions with 100 μm (0.1 mm) increments, with each position measured over five cycles, yielding approximately 250 data points per location (Figure 4d). The minimum separable distance was determined through Hotelling's T^2 test ($\alpha = 0.05$) applied to the four-channel voltage distributions. For each reference position, the test was performed incrementally against neighboring positions on both sides until both left and right distributions were statistically distinguishable from the reference. The distance at which both comparisons passed the significance threshold was recorded as the spatial resolution at that position. Figure 4e presents the channel-wise voltage distributions across the 60 mm trajectory, with mean values and one standard deviation bands for each position. As the contact point progressed from left to right, Channel 1 exhibited a decreasing trend while Channel 3 showed a general increase, reflecting their respective proximity to the electrodes. In contrast, Channels 2 and 4 displayed relatively flat profiles, as the trajectory orientation minimized the distance-dependent voltage modulation for these electrodes. The spatial resolution analysis results are summarized in Figure 4f. The upper panel visualizes the position-dependent minimum separable distance as a color-coded bar, where green indicates the finest resolution (0.1 mm) and red represents the coarsest (0.9 mm) observed within the tested range. The lower panel presents the histogram distribution of these values, revealing a monotonic decrease from 0.1 mm (most frequent) to 0.9 mm. The mean minimum separable distance was determined to be 0.21 mm, confirming sub-millimeter spatial resolution across the sensing area. Notably, the sensor maintained this spatial discrimination at 406 kPa, near its maximum operating pressure. This predictable spatial sensitivity pattern enables reliable position discrimination through the distinct voltage fingerprints generated at different locations, as shown in Figure S12. However, extracting both pressure magnitude and spatial coordinates requires decoupling the coupled information from the four-channel outputs. As the applied pressure increases linearly, the overall voltage levels across all channels increase in a nonlinear manner while maintaining the relative voltage differences between the electrodes, shown in Figure 4b. The observed nonlinear relationship between the four-channel voltage data and the coordinates-pressure data necessitates the use of ML-based decoupling approaches rather than simple linear models. Linear regression models failed to accurately predict spatial positions from the four-channel voltage data, as demonstrated in Figure S13a, confirming that the transformation from voltage signals to position-pressure information cannot be resolved through linear decomposition.

The ML-based approach addresses the inherent nonlinearity of the relationship between the four-channel voltage data and the desired position and pressure outputs. As illustrated in Figure 4g, a multi-layer perceptron (MLP) architecture was

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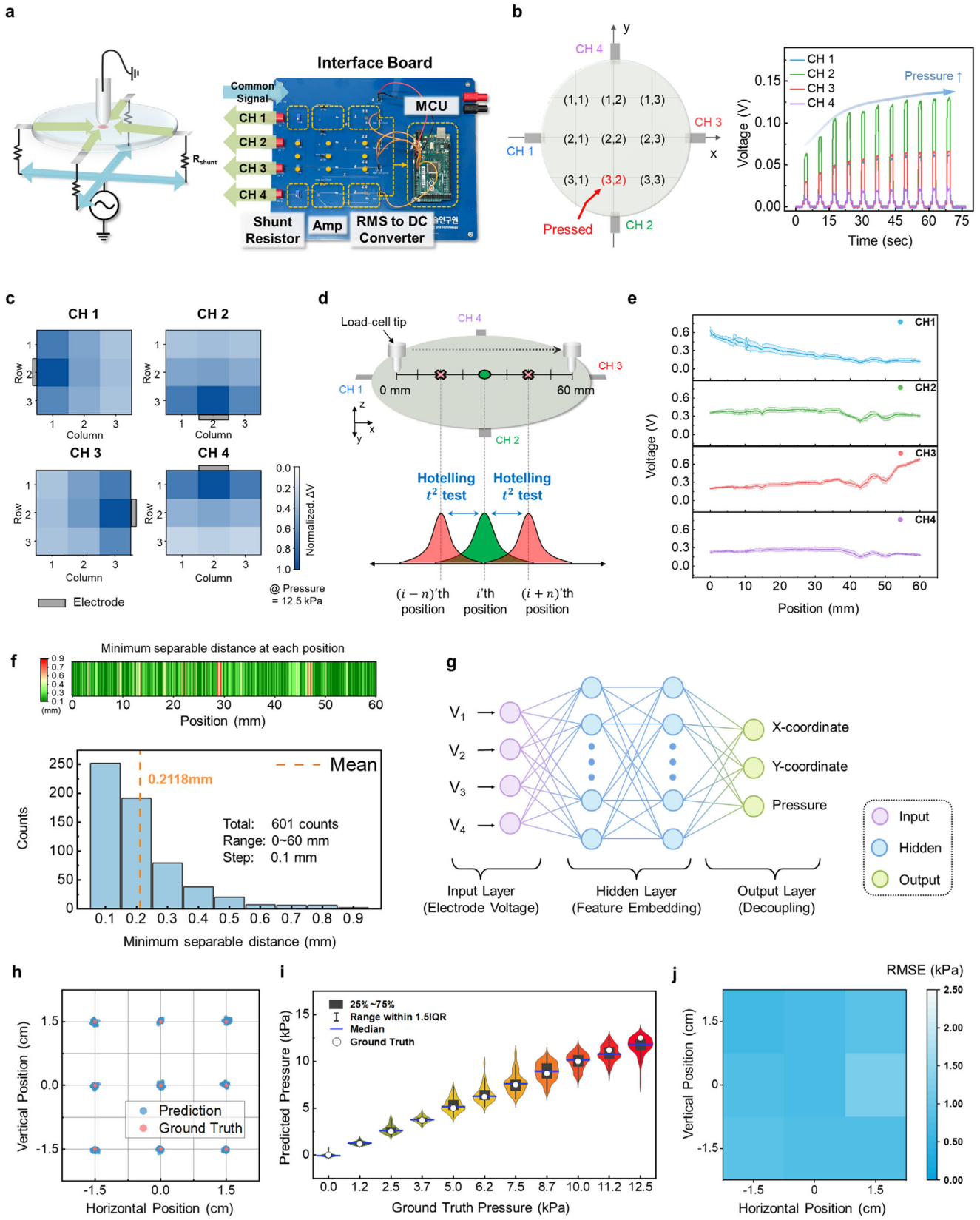

FIGURE 4 | Position and pressure detection via the four-channel iontronic position and pressure sensing system and ML-based analysis. (a) Schematic illustration of the signal processing methodology for the four-channel iontronic pressure sensor system integrated with the interface board, including shunt resistors, amplifiers, RMS-to-DC converters, and an MCU for converting ionic currents to processable voltage signals. (b) Representative voltage measurement of PF-MPS under mechanical stimulus at position (3,2) in the 3 × 3 grid. (c) Channel-wise voltage response maps for each stimulating position in the 3 × 3 grid. (d) Schematic illustration of the experimental setup for spatial resolution measurement, where the load cell systematically pressed along a 60 mm linear trajectory with 100 μm increments. (e) Four-channel voltage distributions along the linear trajectory

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employed for its proven capability as a universal approximator [36, 37, 38], enabling the modeling of complex nonlinear relationships between voltage inputs and the three outputs of spatial coordinates and applied pressure. The network consists of three hidden layers with 64, 128, and 64 nodes, respectively, and rectifying linear unit (ReLU) activation functions, providing sufficient representational capacity while maintaining computational efficiency for real-time applications. Dropout layers with a rate of 0.1 were applied after each hidden layer to prevent overfitting, and early stopping was implemented to halt training if no improvement in test loss was observed for 15 consecutive epochs. Input features were normalized using z-score normalization to ensure stable training. To validate the model performance, comprehensive data acquisition was conducted using the four-channel sensing system illustrated in Figure 4a. The PF-MPS was systematically pressed at nine distinct positions arranged in a 3 × 3 grid configuration with 1.5 cm spacing between adjacent points. At each position, eleven different pressure levels ranging from 0 to 12.5 kPa were repeatedly applied with the load cell, ensuring precise and reproducible mechanical stimulation. The four-channel voltage responses were recorded for each combination of position and pressure, resulting in a dataset comprising 4,056 training samples and 1,014 test samples obtained through an 80:20 train-test split. Position prediction distribution is demonstrated in Figure 4h, where the scatter plot shows predictions (blue points) concentrated near the ground truth (red points) for the nine test grid locations. The MLP successfully decouples position information from the voltage data, achieving an overall root mean square error (RMSE) of 0.0457 cm for position predictions. The concentrated clustering of the predicted values around their corresponding ground truth positions confirms the model's reliable spatial discrimination capability. On the other hand, the overall pressure sensing performance across the full operational range from 0 to 12.48 kPa is analyzed in Figure 4i, where violin and box plots illustrate prediction distributions for eleven discrete pressures. The blue lines represent the median values of predictions, the white circles indicate ground truth values, the colored distributions (green to red) indicate the prediction value distributions, the boxes represent the 25th to 75th percentile range of predictions, and the whiskers indicate the range of predictions excluding outliers. The strong correspondence between prediction medians and ground truth values demonstrates the model's accurate pressure estimation capability. The system achieves an RMSE of 0.198 kPa and a mean absolute error (MAE) of 0.143 kPa for the lowest-pressure label (0 kPa), while the highest-pressure label (12.48 kPa) shows an RMSE of 1.315 kPa. For a scale-independent assessment of prediction accuracy, the relative error is calculated using:

$$\text{Relative Error} = \frac{|\hat{y} - y|}{|y|} \times 100\%$$

where y is the ground truth and ŷ is the prediction. The root mean square of the relative errors across all non-zero pressure levels is 12.9%, while the mean relative error is 9.81%. These performance metrics are particularly significant when compared to human's sensitivity to pressure of approximately 10% [39, 40–42]. The achieved accuracy thus approaches human-like pressure sensitivity, demonstrating the biological relevance and practical applicability of the developed sensing system. Additionally, pressure prediction accuracy across different spatial positions is visualized in Figure 4j through a heatmap of RMSE values for the nine positions, with overall performance achieving RMSE of 0.8445 kPa and MAE of 0.5872 kPa across all grid locations. While some spatial variation in prediction accuracy is observed, the ML-based model demonstrates reliable pressure detection capabilities across all the tested positions.

Following the grid-based validation, character writing and piano keyboard applications were developed to evaluate the model's practical applicability to continuous finger movements and simultaneous multi-touch scenarios. To enhance the model's generalization performance, an additional dataset was collected by drawing five horizontal and five vertical lines with bare fingers following a grid pattern in Figure 5a. The dataset comprised 11,772 training points and 2,943 test points, which provided comprehensive coverage for model development and validation. This experiment was performed according to the regulations of the Institutional Review Board (IRB) of the Korea Institute of Science and Technology (KIST, KIST-202507-HR-002). As shown in Figure 5a, the red trajectories represent ground truth finger movements, while blue traces show the model's point-by-point predictions on test data. The system accurately predicts finger positions from discrete voltage measurements, achieving an RMSE of 0.1315 cm for coordinating prediction and demonstrating millimeter-level accuracy for position prediction applications. Furthermore, PF-MPS successfully decouples position and pressure information in real-time, as demonstrated in Figure 5b, where different pressure intensities are visualized through progressive color-coded representations from green (weak pressure) to red (hard pressure) (Movie S1). The real-time visualization directly demonstrates the effectiveness of the ML-based decoupling approach in simultaneously extracting both spatial and pressure information from the four-channel voltage data. The ML-based position tracking generalizes effectively across various and multiple characters, such as letters (A), numbers (6), and words (SNU, KIST), as illustrated in Figure 5c (Movie S2). The learned feature representations work across different motion trajectories and input styles without requiring application-specific calibration. PF-MPS and the decoupling algorithm operate effectively even in situations different from the controlled training environment. Position and pressure information can be extracted successfully when the

(0–60 mm), indicating the mean values and one standard deviation for each position. (f) Spatial resolution analysis results. Top: position-dependent minimum separable distance represented as a color-coded bar. Bottom: histogram distribution of the minimum separable distances, spanning from the minimum (0.1 mm) to the maximum (0.9 mm) observed values. (g) Schematic illustration of the ML-based decoupling process from the voltage signals to the coordinates and pressure through the MLP. (h) Scatter plot of predicted and ground truth coordinates for nine locations in the 3 × 3 grid. (i) Pressure prediction performance across the full pressure range from 0 to 12.5 kPa, presented as violin and box plots with the x-axis representing the ground truth pressure labels and the y-axis representing the predicted pressure values. (j) Spatial distribution of pressure prediction RMSE across the 3 × 3 grid positions.

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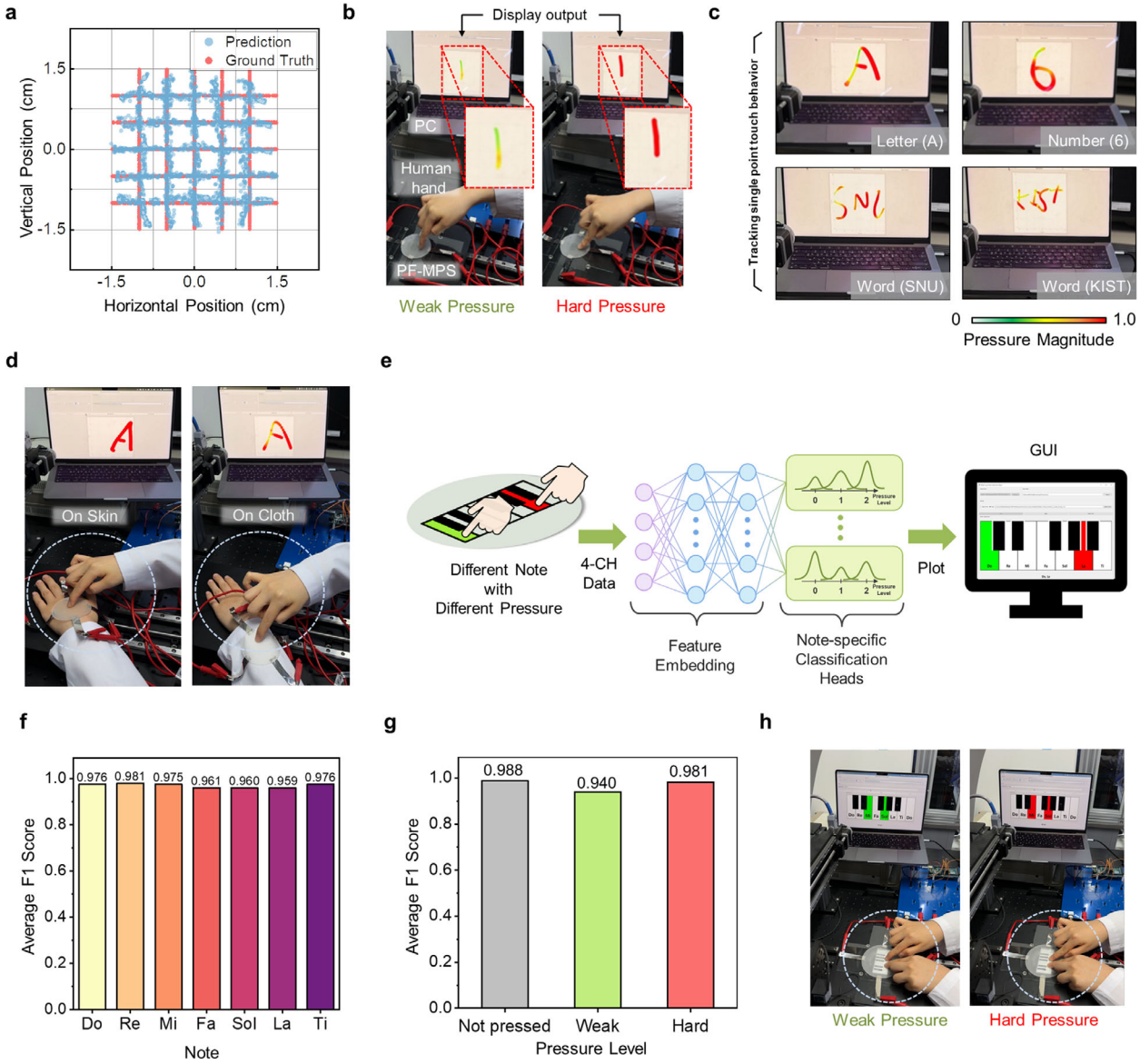

FIGURE 5 | Real-time application demonstrations of the ML-based position and pressure sensing system. (a) Position prediction accuracy for grid pattern drawing: ground truth (red) and predicted (blue) trajectories. (b) Real-time position and pressure decoupling with continuous color coding for pressure levels: weak (left, green-yellow-orange) and hard (right, red). (c) Real-time single-point position tracking demonstration for various character writing. (d) Flexible sensor operation on different surfaces: bare hand (left) and fabric (right). (e) Schematic illustration of multi-touch piano application using MLP with note-specific heads for three-level pressure classification (not pressed, weak, hard) (f) Average F1 scores for individual notes in the piano application. (g) Average F1 scores for the three-class classification (not pressed, weak, hard) across all notes in the piano application. (h) Real-time demonstration of the piano application with simultaneous two-point touch: weak pressure (left) and hard pressure (right).

sensor is placed on bare hands or over fabric surfaces (Figure 5d; Movie S3), demonstrating operational versatility across different contact conditions. In addition, the original MLP model—trained solely on data acquired from a single device under 26°C and 60% RH—remained effective when applied to measurements collected months later from the same device under substantially different environmental conditions (5°C and ~40% RH). Despite the combined influences of temperature/humidity changes and natural material aging, the model successfully reconstructed the writing trajectories with high fidelity (Figure S14), highlighting the strong robustness of both the PF-MPS sensing mechanism and the ML pipeline. Furthermore, when the sensor was touched with a gloved (non-conductive) finger, the signal amplitude was substantially reduced compared to that produced by a bare finger, yet still remained clearly detectable (~tens of mV) (Figure S15). This reduction is attributed to the diminished capacitive coupling effect, as discussed above.

To further examine factors influencing spatial decoding performance, we systematically evaluated the effects of electrode placement and electrode material. We intentionally rearranged the electrodes while keeping the PF-MPS structure and measurement conditions identical, testing (i) a symmetric configuration with four electrodes evenly spaced every 90° and (ii) an asymmetric configuration where opposing electrode pairs were positioned at 150°/30°. As shown in Figure S16a–d, modifying the

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electrode layout substantially altered the electric-field pathways and pressure-induced ion-redistribution behavior, which caused identical pressing coordinates to generate markedly different voltage patterns. Because the ML model was trained using datasets collected under the symmetric 90° arrangement, switching to the asymmetric configuration yielded inconsistent field distributions and altered voltage encoding, resulting in distorted writing trajectories (Figure S16e,f). Additionally, we evaluated the effect of electrode material by replacing stainless steel (SUS) with copper (Cu) while preserving the symmetric 90° geometry (Figure S17a). As shown in Figure S17b, Cu electrodes generated approximately 22% higher voltage amplitude than SUS, which can be attributed to Cu's higher surface charge susceptibility and stronger ion–electron coupling [43]. Importantly, this amplitude increase did not degrade ML-based position tracking, as the MLP model maintained its performance when switching from SUS to Cu electrodes (Figure S17c). This robustness reflects the fact that the model primarily relies on the relative inter-channel voltage relationships rather than absolute amplitudes for spatial decoding.

To validate simultaneous multi-point detection with individual pressure sensitivity for each contact, a piano keyboard application was demonstrated. As illustrated in Figure 5e, data collection was performed by placing a piano-shaped blueprint beneath the PF-MPS and pressing different musical note regions with varying pressure levels (not pressed, weak, strong) using bare fingers. Multiple repetitions were performed for each note-pressure combination to ensure dataset robustness. The collected data is processed through a multi-layer perceptron (MLP) with three shared hidden layers (64, 128, and 64 nodes with ReLU activations and dropout regularization) to extract embedded features from the four-channel voltage inputs. These features are then fed into seven individual classification heads corresponding to each musical note (Do to Ti). Each classification head performs a 3-class classification task using the embedded features to determine whether each note is not pressed, weakly pressed, or strongly pressed. The model is trained using cross-entropy loss, calculated separately for each note and then averaged across all seven outputs. This multi-head architecture enables simultaneous and independent assessment of multiple touch points based on the shared feature representation of the four-channel voltage responses (Figure S13b). The classification performance was evaluated using F1 scores calculated individually for each musical note, which provide a balanced measure of precision and recall calculated through the following equation:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The piano keyboard application achieves high classification accuracy across all musical notes. Performance ranges from 0.959 to 0.981 across different musical notes, which demonstrates consistent accuracy throughout the keyboard interface (Figure 5f). Average F1 scores for the three-class classification (not pressed, weak, hard) across all notes are presented in Figure 5g, achieving high accuracy with F1 scores of 0.988 for not pressed, 0.940 for weak pressure, and 0.981 for hard pressure states. These results demonstrate the system's ability to reliably distinguish between different interaction states with high precision. Similar to the handwriting application, real-time operation of the multi-touch piano keyboard application is shown in Figure 5h. A real-time demonstration of the PF-MPS device performing "The Celebrated Chop Waltz" on the multi-touch piano keyboard is provided in Movie S4. The system confirms its capability for simultaneous multi-point pressure-sensitive detection in practical use scenarios. This multi-point functionality enables complex interactive applications that require simultaneous monitoring of multiple contact points with individual pressure sensitivity. The integration of spatial accuracy, pressure discrimination, and multi-touch capabilities establishes the developed system as a versatile platform for human-machine interface applications requiring both precision and intuitive operation.

3 | Conclusion

In summary, we developed a pixelation-free, monolithic iontronic pressure sensor featuring a minimal lateral electrode design for simultaneous pressure and position sensing. Unlike conventional vertical sandwich-structured sensors that require complex sensor arrays and dense wiring, PF-MPS operates with only 2 or 4 bottom electrodes, enabling both structural simplicity and multifunctionality. The sensor demonstrates a high sensitivity of 4.16 kPa^{-1}, a wide detection range up to 455 kPa, and excellent durability under repetitive loading and long-term stability. Notably, PF-MPS maintains stable signals even under static pressure for extended durations (>3300 s), overcoming a key limitation of existing iontronic sensors. Mechanism studies with synchrotron XRD and EIS confirmed that pressure-induced ion release and redistribution within the ionogel govern the electrical response. Sensor performance was further optimized through control of ionic liquid content and AC driving parameters. Furthermore, a machine learning-based model enabled accurate decoupling of pressure and position using only four-channel voltage outputs. Real-time applications, including handwriting tracking and piano interface, even on curved surfaces such as human skin and textiles, validated the system's robustness and versatility. Overall, this work presents a scalable and structurally simple pressure-sensing platform that operates fast enough for real-time HMI demonstrations and is well-suited for next-generation e-skin, wearable electronics, and interactive interfaces, with further optimization potentially extending its applicability to soft robotics.

4 | Experimental Section

4.1 | Materials

Poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP, Mw ≈ 400 000), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][TFSI], ≥98%), and acetone (99.9%) were all purchased from Sigma–Aldrich and used as received without further purification.

4.2 | Preparation of Ionogel Film for PF-MPS

PVDF-HFP was first dissolved in acetone solvent at a mass ratio of 1:6.272 (PVDF-HFP: acetone) by stirring at 60°C for 2 h. Subsequently, [EMIM][TFSI] ionic liquid was added to the solution at

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a concentration of 20 wt.%. The mixture was stirred continuously at 60°C for 24 h to ensure complete homogenization. The final solution was then cast into a glass petri dish and dried at room temperature until the solvent fully evaporated, forming the ionogel sensing layer.

4.3 | Fabrication of PF-MPS

Stainless steel (SUS) films (thickness = 80 μm) were cut into rectangular strips with a width of 1 cm and used as electrodes and placed beneath the ionogel film. For the two-electrode layout, the electrodes were positioned 180° apart with respect to the center of the ionogel film. In the four-electrode configuration, the electrodes were arranged at 90° intervals, radially oriented toward the center of the ionogel film.

4.4 | Evaluation of Pressure and Spatial Sensing Performance

An arbitrary function generator (AFG3152C, Tektronix) was used to apply an AC signal to all electrodes. A motorized XYZ stage (Namil Optical Instruments Co., Korea) equipped with a load cell (CDFSB-10, Bongshin Loadcell Co., Korea) and interchangeable SUS and POM tips (Namil Optical Instruments Co., Korea) was employed to apply and measure pressure. When using two electrodes, the current was measured by connecting a current meter (Multimeter, Model DAQ6510, Keithley) to the target electrode. For 1D position sensing, pressure was applied in a straight line along either the *x*- or the *y*-axis at regular intervals. A custom-designed PCB board (JLCPCB) enabled simultaneous signal acquisition and visualization across all channels using a computer interface. For current measurements, a shunt resistor was connected in series with each electrode, and the voltage drop across the resistor was measured to calculate the current. These small voltage signals were amplified using a current sense amplifier and then processed by an RMS-to-DC conversion circuit to obtain the RMS values of the AC signal. The resulting DC voltage was fed into the analog input ports of an Arduino Mega microcontroller for digital conversion and processing. For the machine-learning process, the dataset comprised 11772 training points and 2,943 test points was collected by drawing five horizontal and five vertical lines with the participant's bare finger. To confirm the environmental robustness of the system, both the voltage signal generation and press-position tracking tests were performed under two different conditions (Temp. 26°C, RH~60% and Temp. 5°C and RH ~40%). This experiment was conducted in accordance with the regulations of the Institutional Review Board (IRB) of the Korea Institute of Science and Technology (KIST, approval number: KIST-202507-HR-002).

4.5 | Electrochemical Characterization

Electrochemical impedance spectroscopy (EIS) was performed using a Metrohm Autolab analyzer with NOVA software to analyze electrochemical impedance behavior under pressure, applying an AC voltage over a frequency range of 0.1 Hz to 500 kHz. Pressure was applied using a mechanical pushing machine (SNM, Korea) during EIS measurements. The influence of contacting material on signal response was examined using tips made of SUS and plastic during measurements.

4.6 | Synchrotron XRD Analysis for Pressure-Induced Changes

To analyze structural changes under mechanical loading, XRD measurements were performed in transmission mode at the 3C-SAXS beamline of the Pohang Accelerator Laboratory in Korea. All ionogel samples had a thickness of 1 mm. A sample holder was utilized to immobilize the ionogel films during analysis. Before applying pressure, XRD patterns were acquired in the absence of external pressure by directing the X-ray beam through the unpressed ionogel. Subsequently, a uniform mechanical pressure was applied to the sample using a customized experimental setup (Figure S3). The applied pressure (σ) exerted on the ionogel film was estimated to be 354 kPa, with the value estimated from the measured thickness change (Δt) and the measured Young's modulus (E, 18.3 MPa) using the linear elastic relation $\sigma = E(\Delta t/t)$. Spectral features corresponding to the PI films were subtracted from the raw data before interpretation to prevent interference from the crystalline peaks of PI.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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